



Winning Space Race with Data Science

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Outline

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Executive Summary

Summary of methodologies:

- Data from the SpaceX Rest API and Wikipedia records were collected, cleaned, and prepped
- Exploratory data analysis alongside data visualization and feature engineering were performed to get a preliminary picture of the data
- Using interactive visualizations such as dashboards and folium, the data was explored further
- Finally, a machine learning model, using different classification methods, was built and evaluated

Summary of results:

- Results from the explanatory data analysis revealed which features are better predictors of successful launches
- Interactive dashboards displayed how payload mass and launch site influence launch success, while mapping with folium displayed the key factors such as area, distance, and surroundings of the launch sites
- Predictive analysis using model building and evaluation identified the decision tree classifier as the best machine learning model for making these predictions

Introduction

Project background and context:

In the commercial space age, space travel is becoming more affordable. Space X is a leading company in this industry that leverages the first stage of rocket launches, specifically the Falcon 9, to make space travel affordable. The aim is to make Space Y a competitive company by determining the price of each launch. In order to achieve this, information from Space X rocket launches will be gathered to train a machine learning model that will predict whether the first stage of the rocket launch will be successful or not.

Problem Questions:

- How does the relationship between variables such as, Pay Load Mass, Launch Site, and Orbit Type, among others, affect the launch outcome?
- Where are launch sites situated and what is their impact on landing success?
- Which learning model can accurately predict the landing outcome of the Falcon 9?

Section 1

Methodology



Methodology

Executive Summary

Data collection methodology:

- Data on Falcon 9 launches was collected using the Space X REST API and web scrapping of Wikipedia records

Performed data wrangling:

- Data was cleaned and prepped by filtering the data to include only Falcon 9 launches, handling missing values, and assigning training labels to mission outcomes

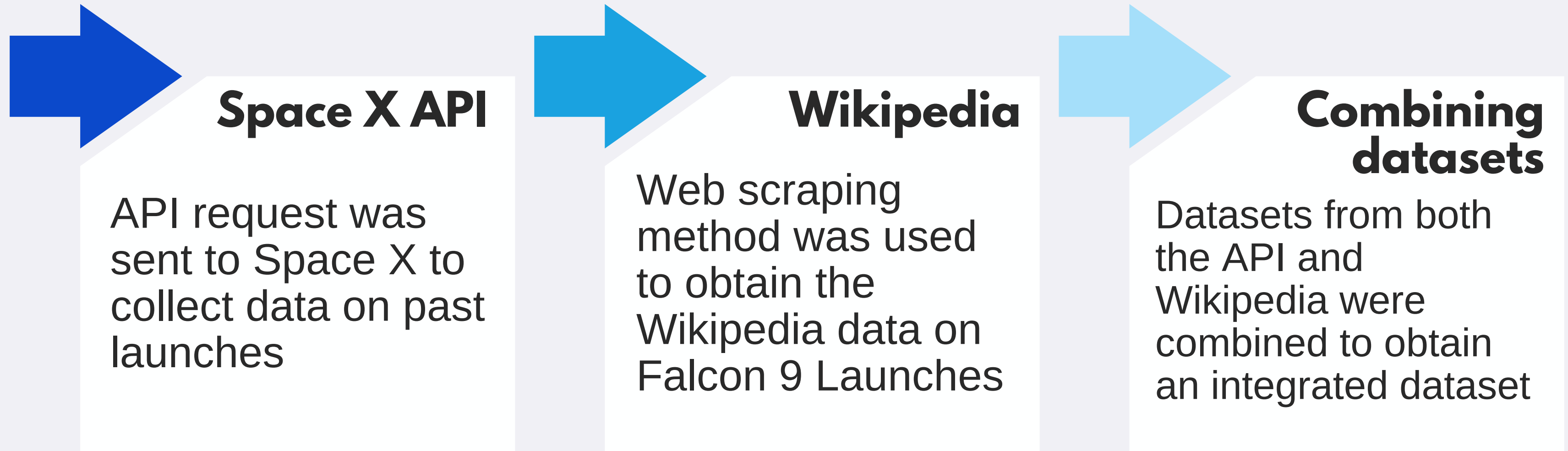
Performed exploratory data analysis (EDA) using visualization and SQL

Performed interactive visual analytics using Folium and Plotly Dash

Performed predictive analysis using classification models

- Classification models were built, tuned, and evaluated to determine the best performing model

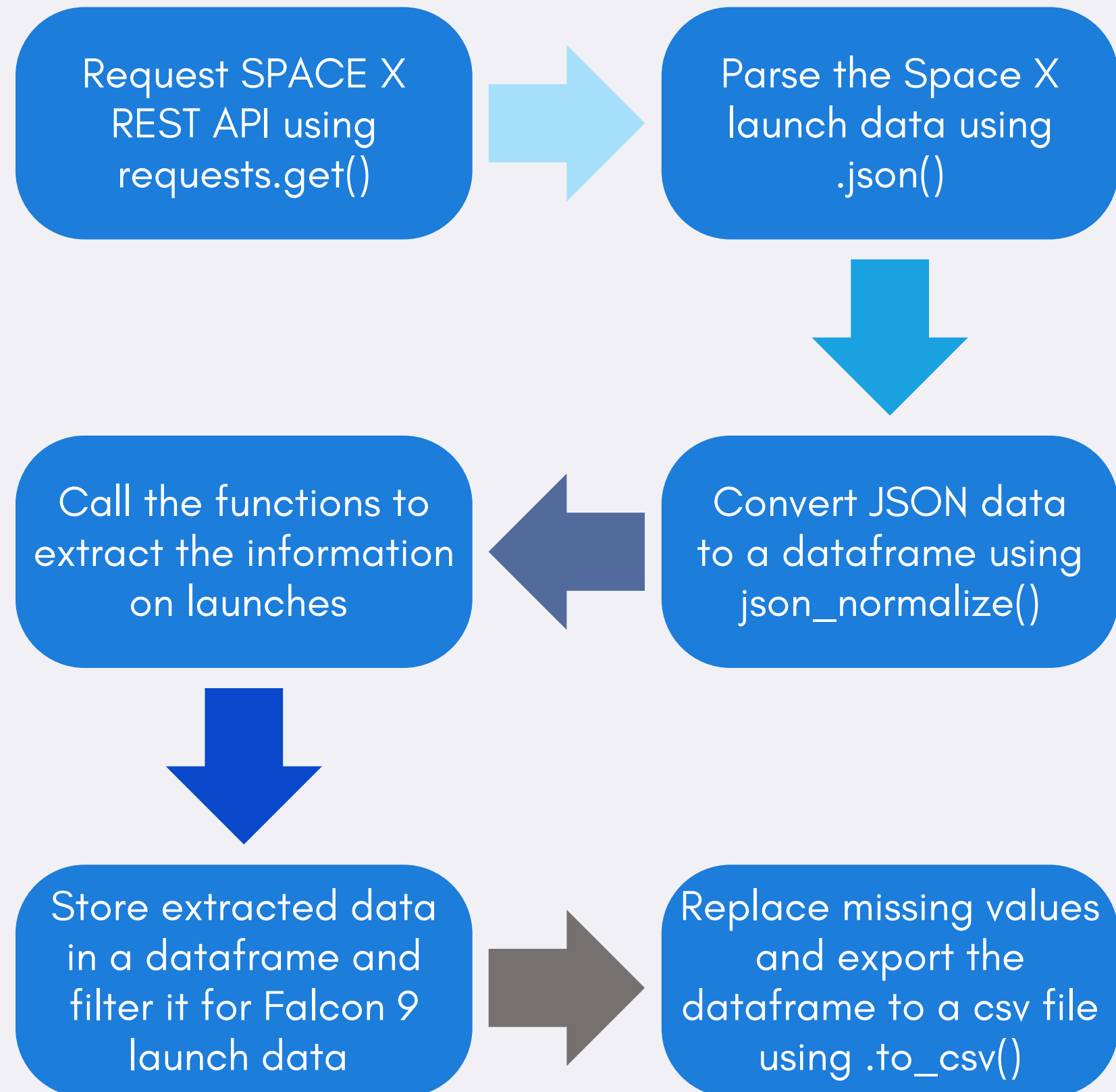
Data Collection



Data Collection – SpaceX API

Data from Space X REST API was requested, parsed, and converted to a dataframe.

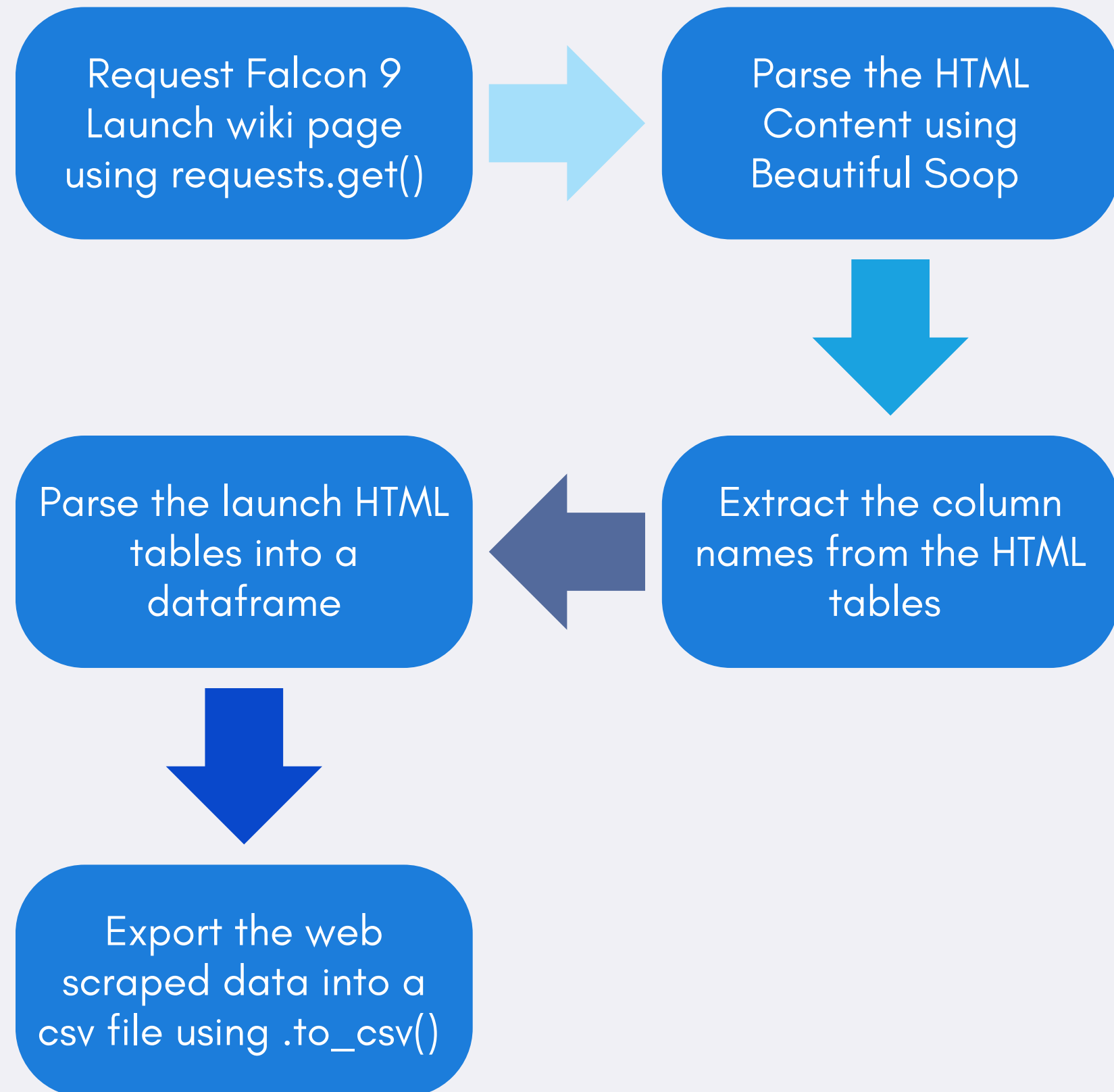
[GitHub Link](#)



Data Collection – Scrapping

Data from Wikipedia page was collected via web scraping, parsed, and converted into a dataframe

[GitHub Link](#)



Data Wrangling

Data wrangling was performed by categorizing the different mission outcomes into specified training labels (0 or 1).

[GitHub Link](#)

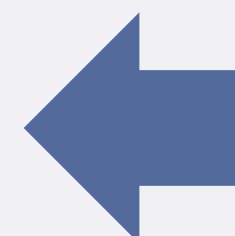
Perform EDA on the data to identify null values and data types



Calculate the number of launches on each site



Calculate the number and occurrence of each orbit type



Create training labels (1, 0) and assign mission outcomes accordingly



Integrate the new column into the dataframe and export it using .to_csv

EDA with Data Visualization

Exploratory Data Analysis (EDA) was performed to determine which attributes were correlated with successful launches. To uncover these relationships, plots were generated such as:

Scatterplots

- These charts were plotted to identify the relationship between the attributes:
 - FlightNumber and PayloadMass
 - FlightNumber and LaunchSite
 - Payload Mass and Launch Site
 - FlightNumber and Orbit type
 - Payload Mass and Orbit type

Barplot

- This plot was generated to visualize the success rate for each orbit type

EDA with Data Visualization

Exploratory Data Analysis (EDA) was performed to determine which attributes were correlated with successful launches. To uncover these relationships, plots were generated such as:

Lineplot

- This plot was generated to visualize the yearly trend of successful launches

Features Engineering was then performed by analyzing which features revealed to be more correlated with the success rate of each launch

- Categorical data was transformed into numerical data using `pd.get_dummies()` and casting the entire dataframe into float data type

EDA with SQL

The following SQL queries were performed:

- Displaying the names of the unique launch sites in the space mission
- Displaying 5 records where launch sites begin with the string 'CCA'
- Displaying the total payload mass carried by boosters launched by NASA (CRS)
- Displaying average payload mass carried by booster version F9 v1.1
- Listing the date when the first successful landing outcome in ground pad was achieved
- Listing the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- Listing the total number of successful and failure mission outcomes

EDA with SQL

The following SQL queries were performed:

- Listing all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function
- Listing the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015
- Ranking the count of landing outcomes (such as Failure (drone ship) or Success (ground pad) between the date 2010-06-04 and 2017-03-20, in descending order

Build an Interactive Map with Folium

The following map objects were created with folium:

Markers

- Created to mark launch sites on the map

Circles, Pop-up label, Text label

- Created to highlight the area around the launch sites and added pop-up and text labels to indicate their names

Colored Markers

- Created to distinguish between successful (green) launch site coordinates and failed (red) launch site coordinates

Lines

- Created to map out the distance between two coordinates, specifically the distance between a launch site and its nearest proximities

Build a Dashboard with Plotly Dash

The following plots/graphs were added to the dashboard:

Pie Chart

- This graph displayed the percentage of successful launches for all sites as well as individual launch sites

Scatter Plot

- This graph displayed the relationship between payload mass and launch site for both all sites and individual sites

To make the dashboard more interactive, these input components were also added:

Dropdown List

- This allows for the selection of specific or all launch sites

Range Slider

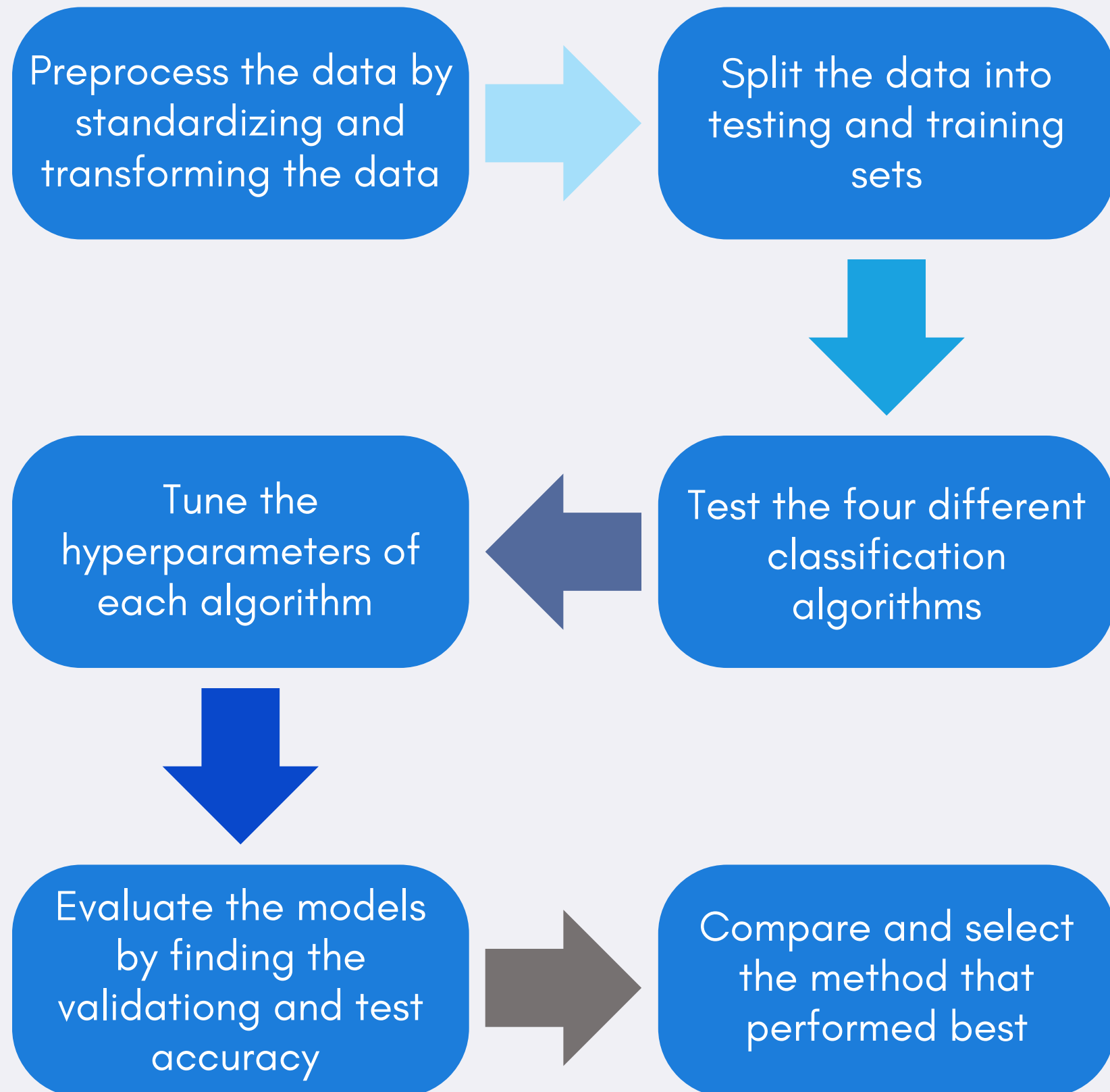
- This allows for the selection of a specific range for the payload mass

Predictive Analysis (Classification)

Predictive analysis was carried out by testing four Classification algorithms:

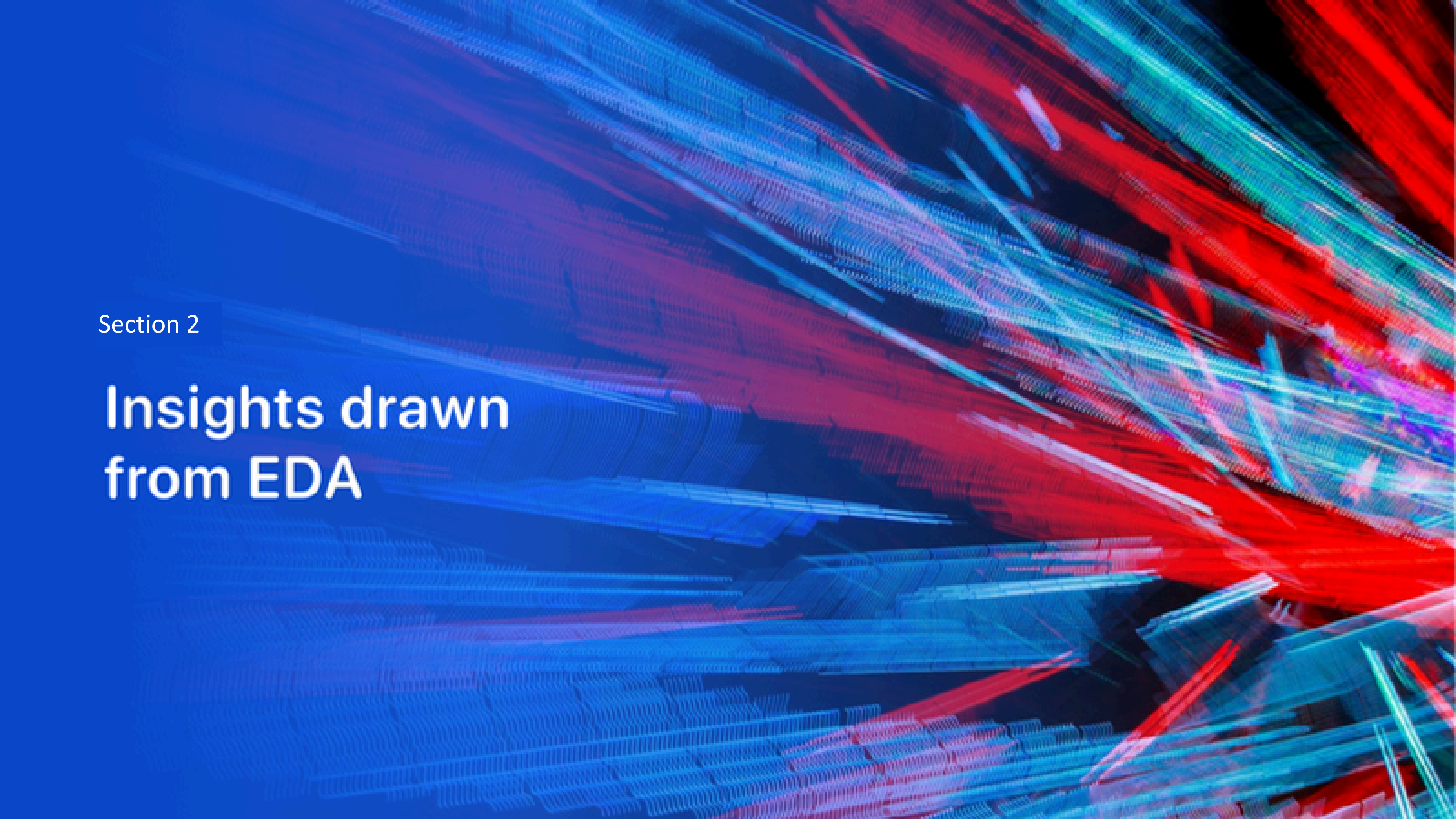
- **Logistic Regression**
- **Support Vector Machine**
- **Decision Tree Classifier**
- **K Nearest Neighbors**

[GitHub Link](#)



Results

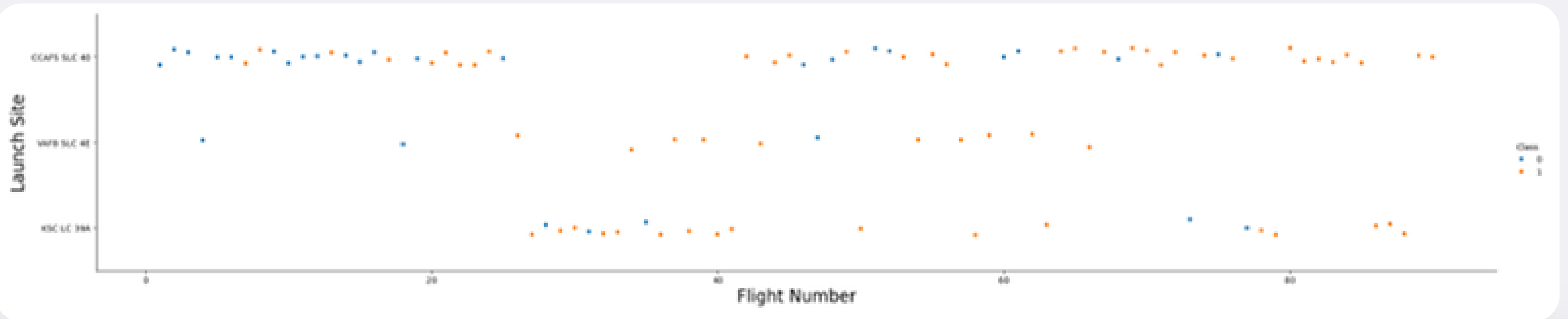
- **Exploratory data analysis results**
- **Interactive analytics demo in screenshots**
- **Predictive analysis results**

The background is a complex, abstract composition. It features a dark blue base with numerous bright blue and red streaks, lines, and brushstrokes radiating across the frame. A faint, light blue grid pattern is visible, particularly on the left side. The overall effect is one of dynamic energy and digital complexity.

Section 2

Insights drawn from EDA

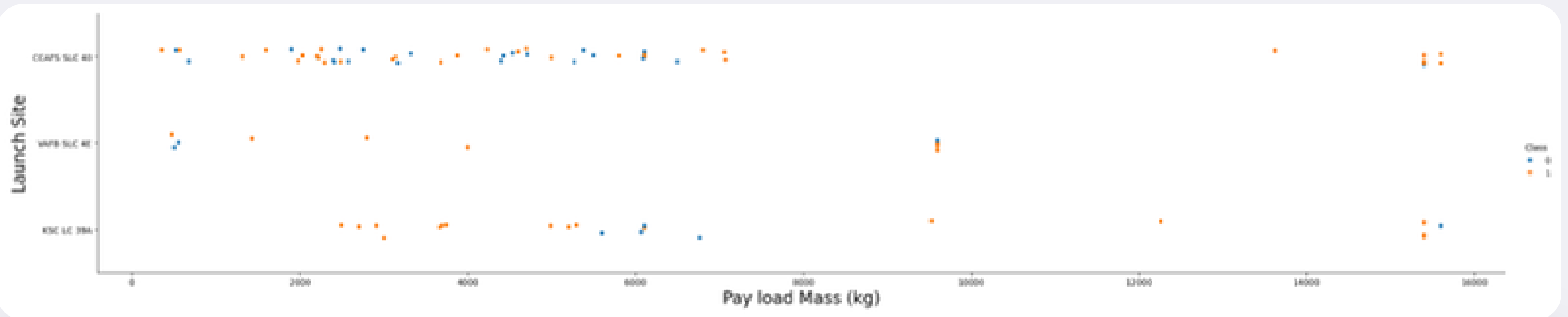
Flight Number vs. Launch Site



Inference:

- As the number of flights increased, the rate of success also increased for the launch sites
- Among the launch sites, CCAFL SLC 40 has the highest number of flights as well as most successful launches
- While VAFB SLC 4E has the least number of flights and successful launches, followed by KSC LC 39A

Payload vs. Launch Site



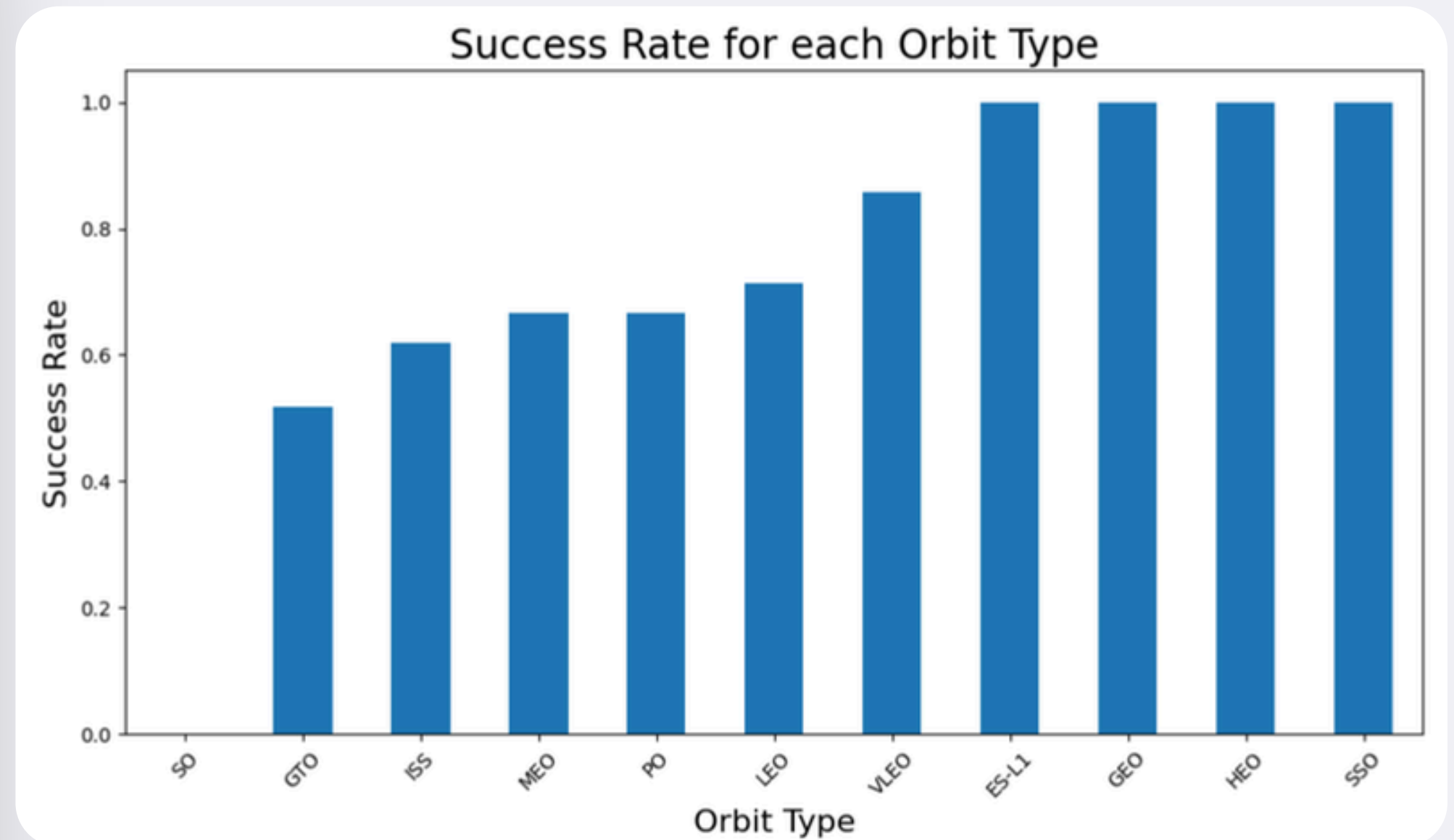
Inference:

- Most of the launches happened with a payload mass below 7000 kg
- There were sparse launches with a higher payload mass, above 7000 kg, but they were mostly successful
- The launch site KSC LC 39A had all successful launches for pay load mass under 5500 kg

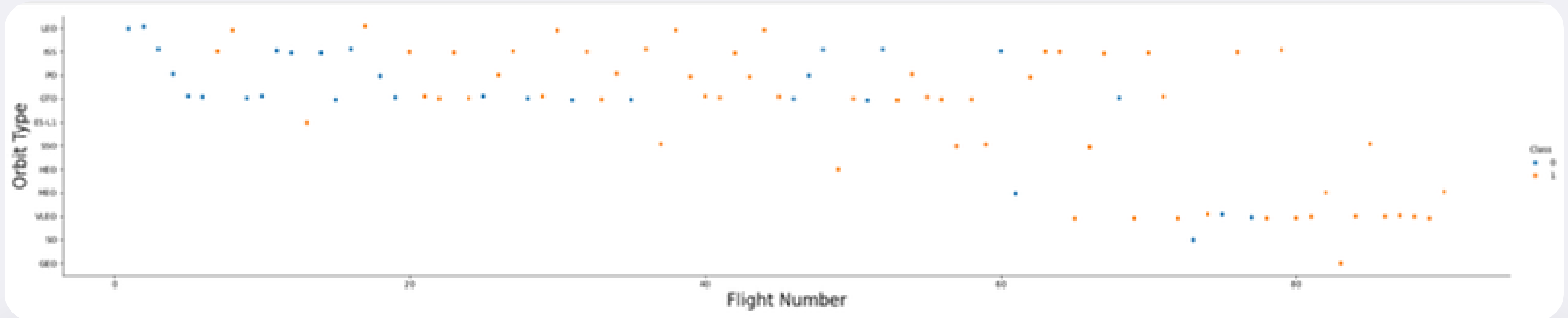
Success Rate vs. Orbit Type

Inference:

- The success rate according to orbit type is **lower** for the orbits **GTO, ISS, MEO, PO, LEO, and VLEO**, indicating that launch missions in these orbits may involve uncertainties.
- While the success rate according to orbit type is **highest** for the orbits **VLEO, ES-L1, GEO, HEO, SSO**, indicating that launch missions in these orbits may be more reliable and certain.



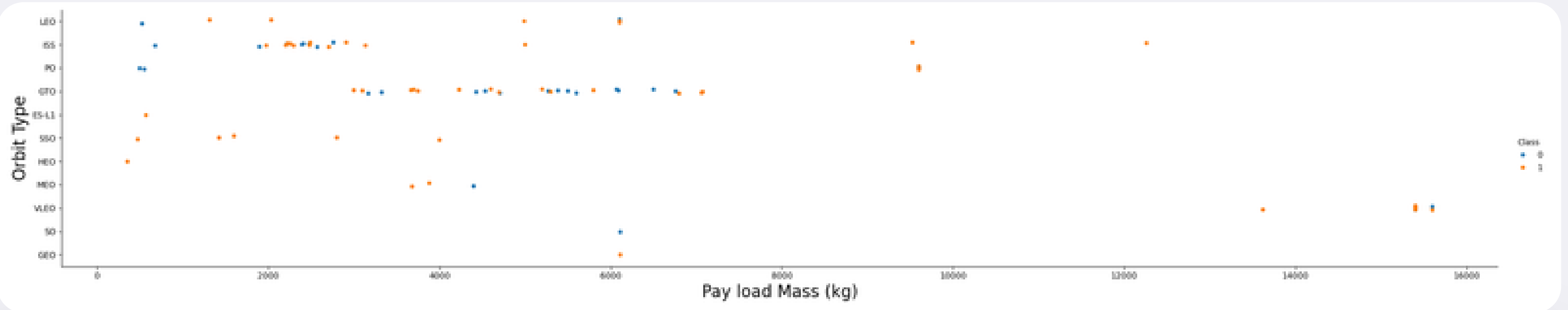
Flight Number vs. Orbit Type



Inference:

- Earlier launches have mostly been done in the orbits GTO, PO, ISS, and LEO
- As the number of launches increased, their rate of success also improved accordingly
- While the orbits VLEO, ES-L1, GEO, HEO, SSO have higher launch success rates, it is to be noted that they have had the least number of flights as compared to the other orbits

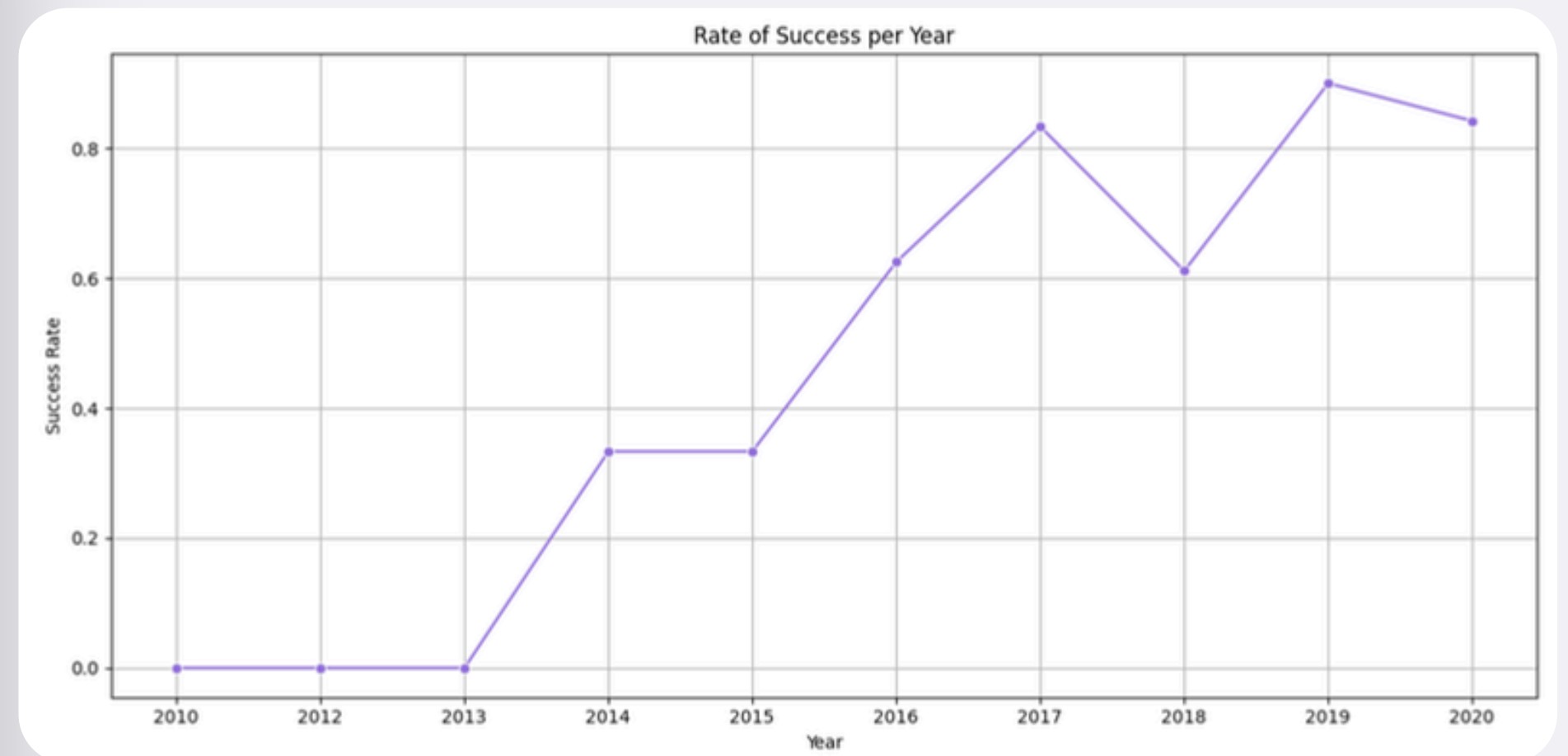
Payload vs. Orbit Type



Launch Success Yearly Trend

Inference:

- The line plot shows that after 2013, the success rate of the launches has kept on increasing
- Although there was a descend in the success rate in the year 2018, the overall trend shows that success rate has been increasing
- The year 2019 had the highest rate of successful launches while the years 2010, 2012, and 2013 remained unsuccessful



All Launch Site Names

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Inference:

- There are four distinct launch sites in the dataset

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.dbDone.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Inference:

- This table displays the records of five launch sites whose names begins with 'CCA'

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total_Payload_Mass

45596

Inference:

- This table displays the total pay load mass in kg carried by the boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

```
❯sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
AVG(PAYLOAD_MASS__KG_)
```

```
2928.4
```

Inference:

- This table displays the average payload mass carried by the booster version F9 v1.1

First Successful Ground Landing Date

```
sqlite> SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

MIN(Date)

2015-12-22

Inference:

- This table displays the date when the first successful landing outcome in ground pad was achieved

Successful Drone Ship Landing with Payload between 4000 and 6000

```
sqlite> SELECT DISTINCT Booster_Version FROM SPACE_TABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Inference:

- This table displays the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

```
\sql SELECT Mission_Outcome, COUNT (*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Inference:

- This table displays the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

```
\sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

Inference:

- This table displays the booster versions that have carried the maximum payload mass

2015 Launch Records

Inference:

- This table displays the records with the month names, failure landing outcomes in drone ship ,booster versions, and launch site for the months in year 2015

```
sqlite
SELECT
  CASE substr(Date, 6,2)
  WHEN '01' THEN 'January'
  WHEN '02' THEN 'February'
  WHEN '03' THEN 'March'
  WHEN '04' THEN 'April'
  WHEN '05' THEN 'May'
  WHEN '06' THEN 'June'
  WHEN '07' THEN 'July'
  WHEN '08' THEN 'August'
  WHEN '09' THEN 'September'
  WHEN '10' THEN 'October'
  WHEN '11' THEN 'November'
  WHEN '12' THEN 'December'
  END AS Month,
  Landing_Outcome,
  Booster_Version,
  Launch_Site
FROM SPACEXTABLE
WHERE substr(Date,0,5)='2015'
  AND Landing_Outcome = 'Failure (drone ship)';
```

* sqlite:///my_data1.db
Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Inference:

- This table displays the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

```
mysql SELECT Landing_Outcome,
COUNT (*)
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
Landing_Outcome
ORDER BY
COUNT (*) DESC;
```

* sqlite:///my_data1.db
Done.

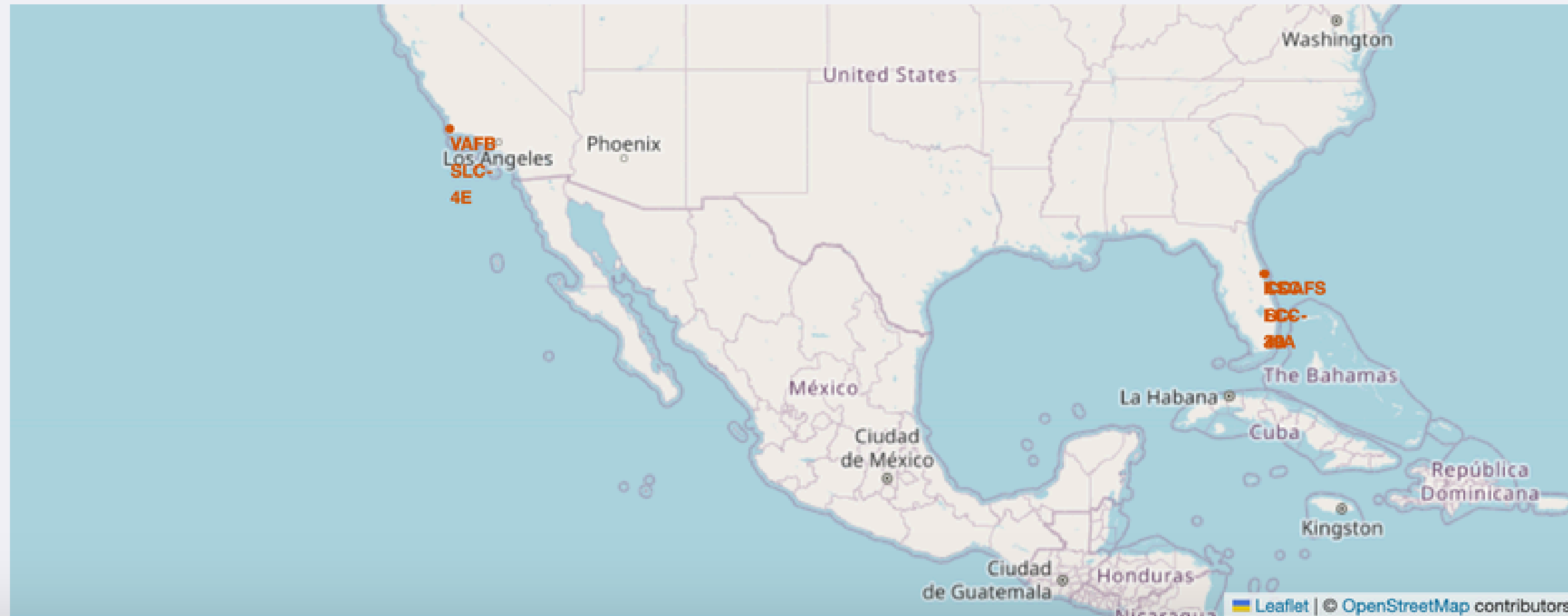
Landing_Outcome	COUNT (*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and a dense network of city lights at night. The lights are concentrated in the lower right quadrant, while the upper left shows the dark blue of the atmosphere and the blackness of space.

Section 3

Launch Sites Proximities Analysis

Launch Site Locations



Inference:

- The launch sites are above the equator line and are in close proximity to the coast.
- These positions could be intentional due to physics and can serve as safety measures when implementing these rocket launches.

Marked and Clustered Launch Outcomes

Inference:

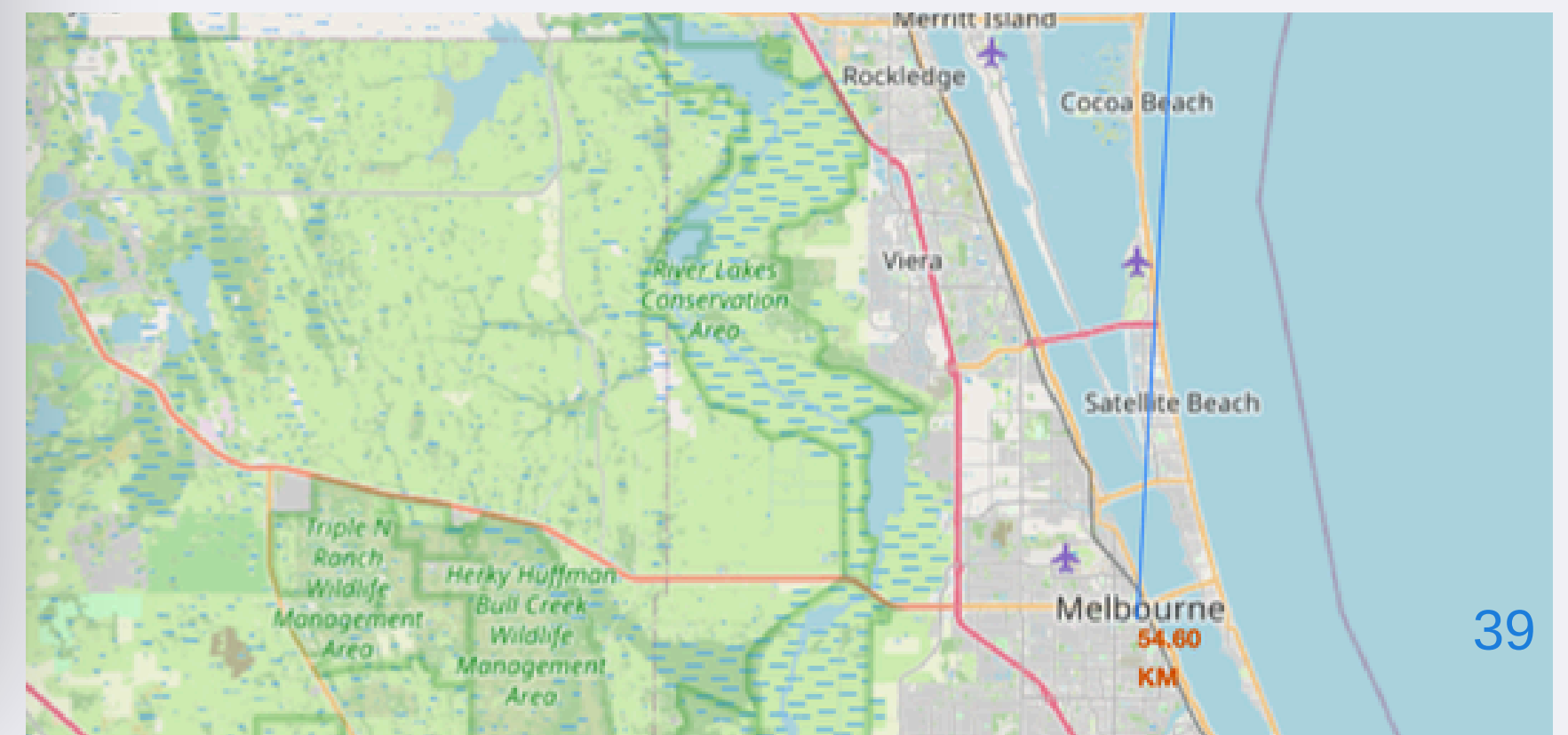
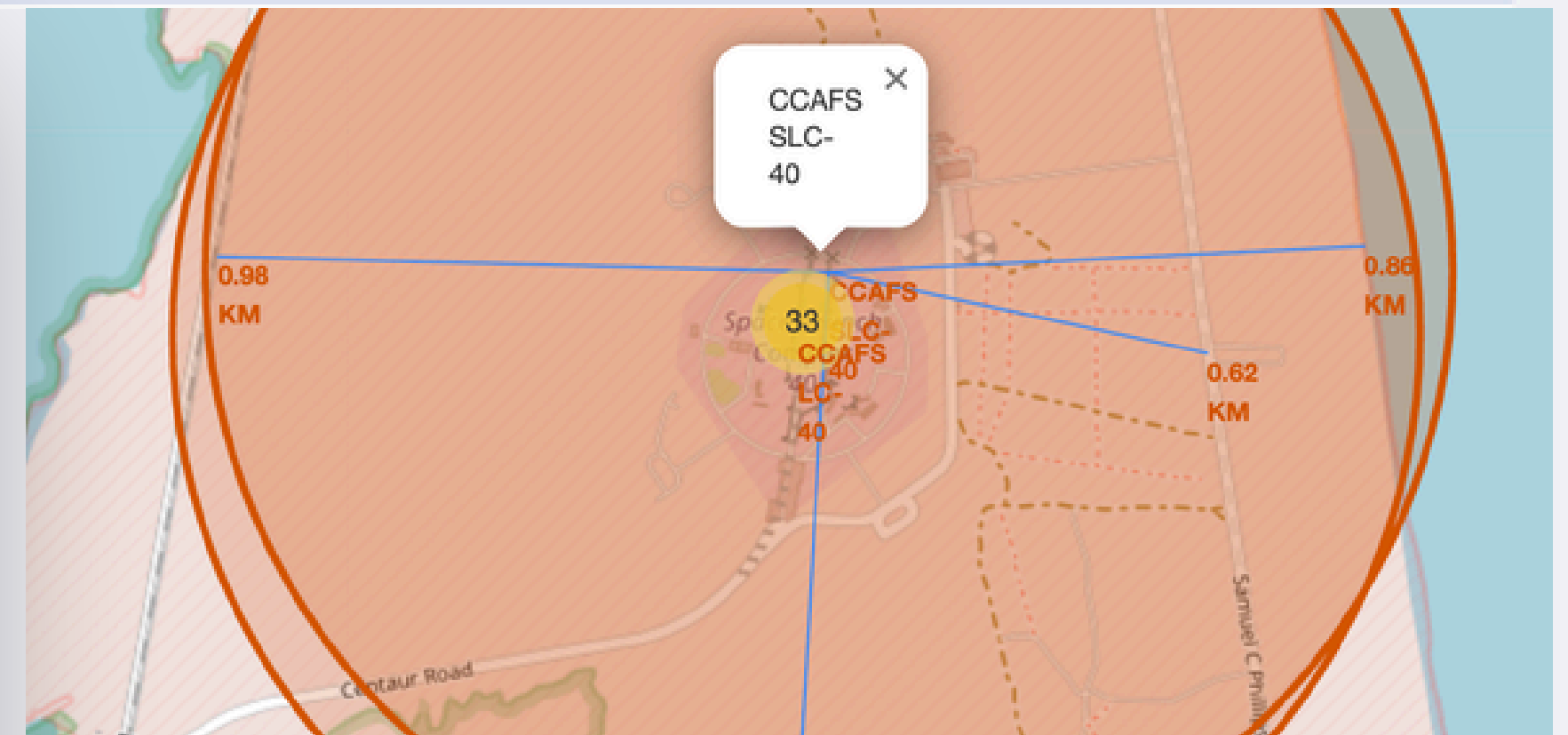
- Clustering the markers aided in a better visualization of the launch sites
- Adding colored markers made it more efficient to distinguish between successful and unsuccessful launch outcomes depending on the site
- The pop-up label was instrumental in identifying the specific launch site being explored

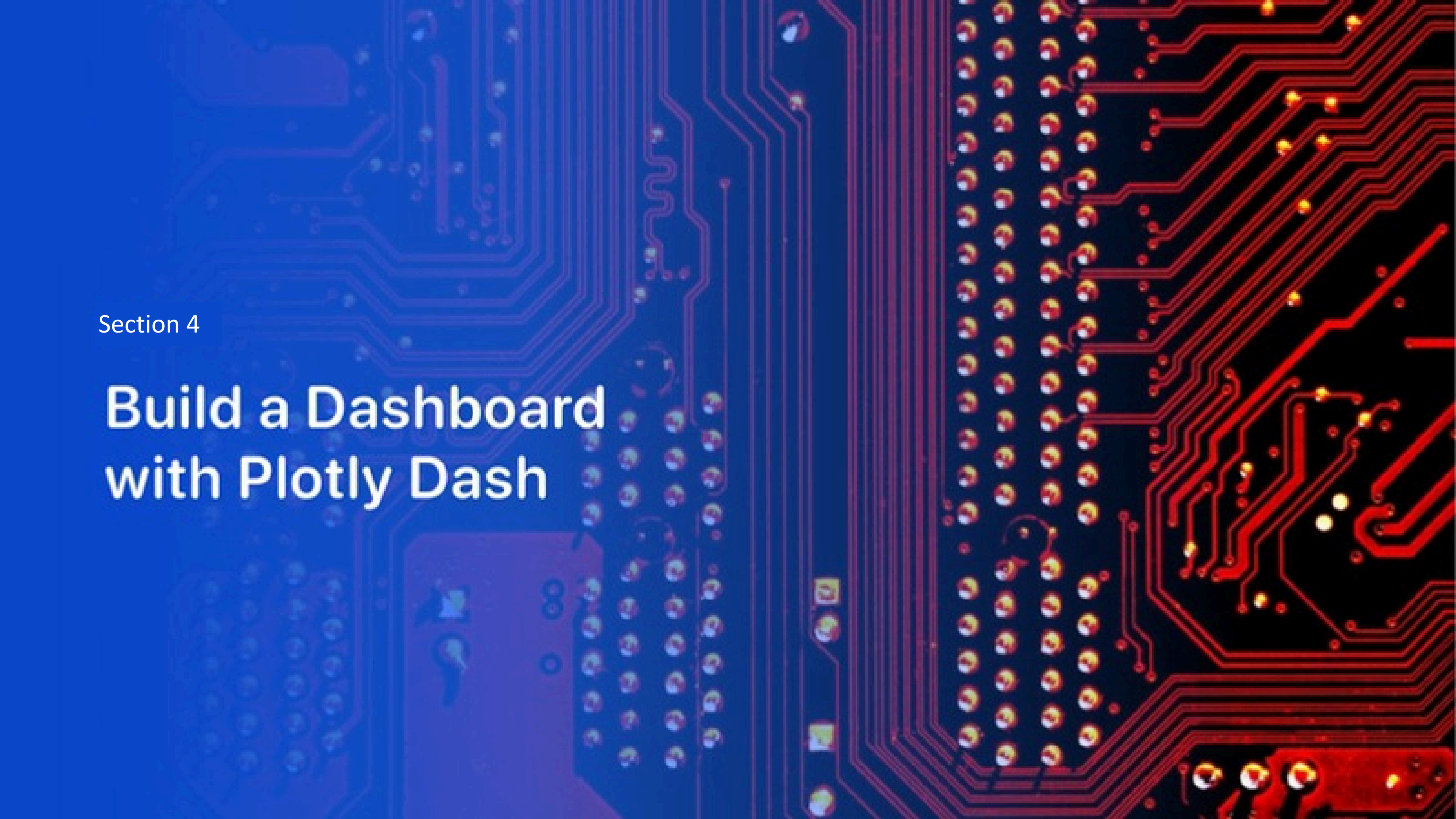


Launch Site Proximities

Inference:

- While launch sites differ in their proximities, here the launch site CCAFS SLC-40 has the coastline, railway, highway, and city near its vicinity
- It can be analyzed that a highway, coastline, and railway are relatively closer in proximity to the launch site.
- With the highway being the closest with a calculated distance of 0.62 km.
- The city is farthest away from the launch site with a calculated distance of 54.60 km.
- It is logistically sound for a launch site to be far from populated areas as compared to highways, and railways near the coastline

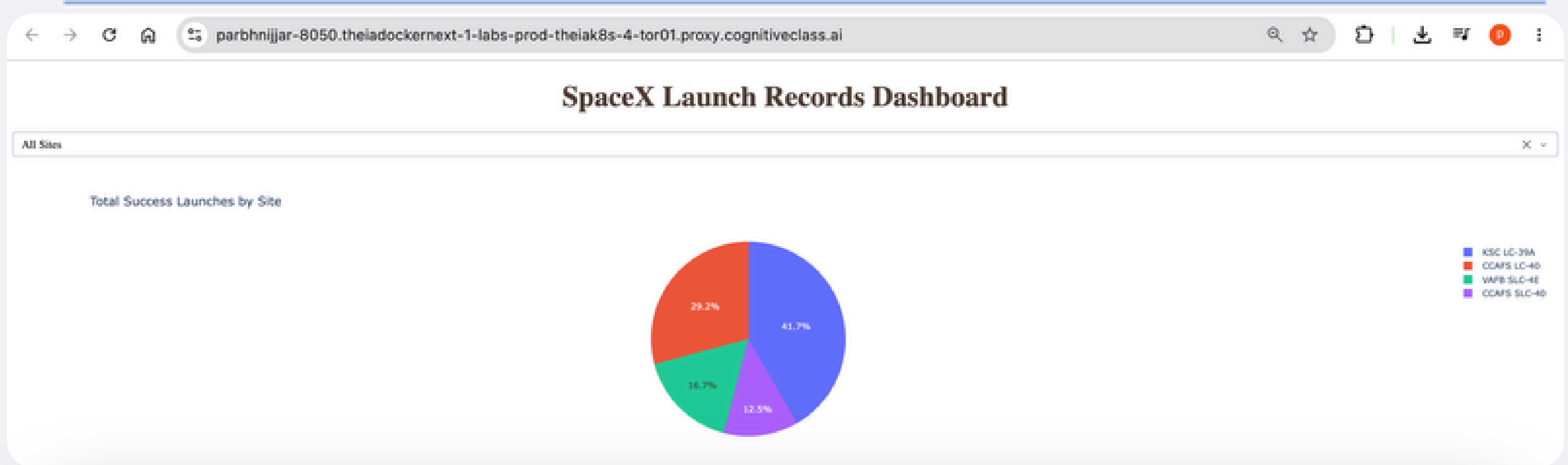




Section 4

Build a Dashboard with Plotly Dash

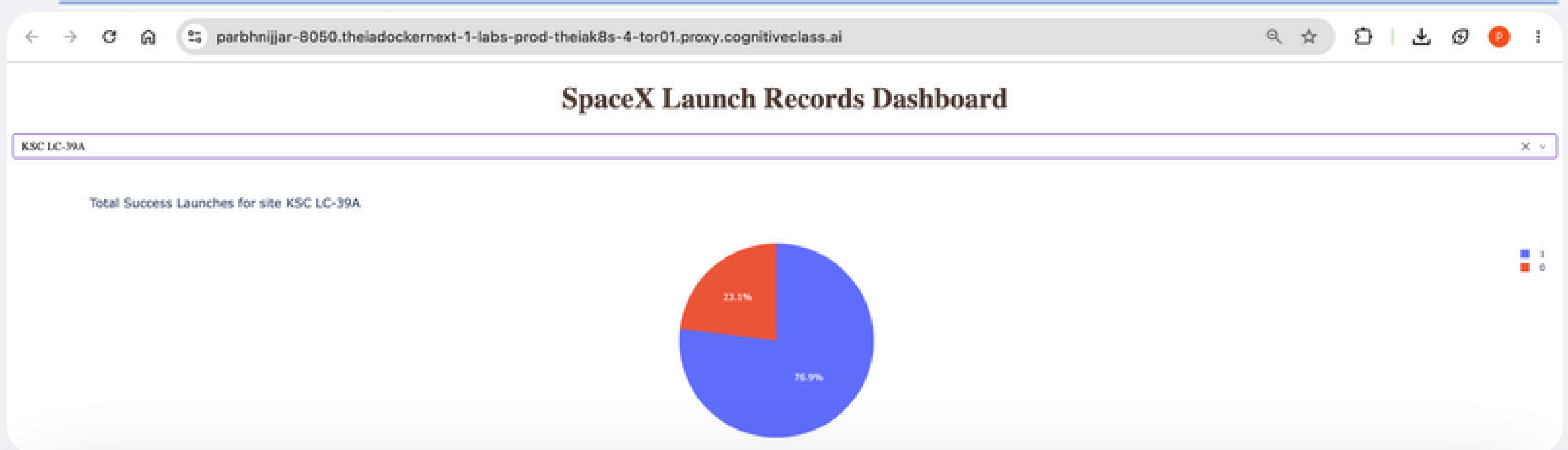
Total Success Launches by Site



Inference:

- The launch site KSC LC-39A has the highest total for successful launches.

Launch Success Rate for KSC LC-39A



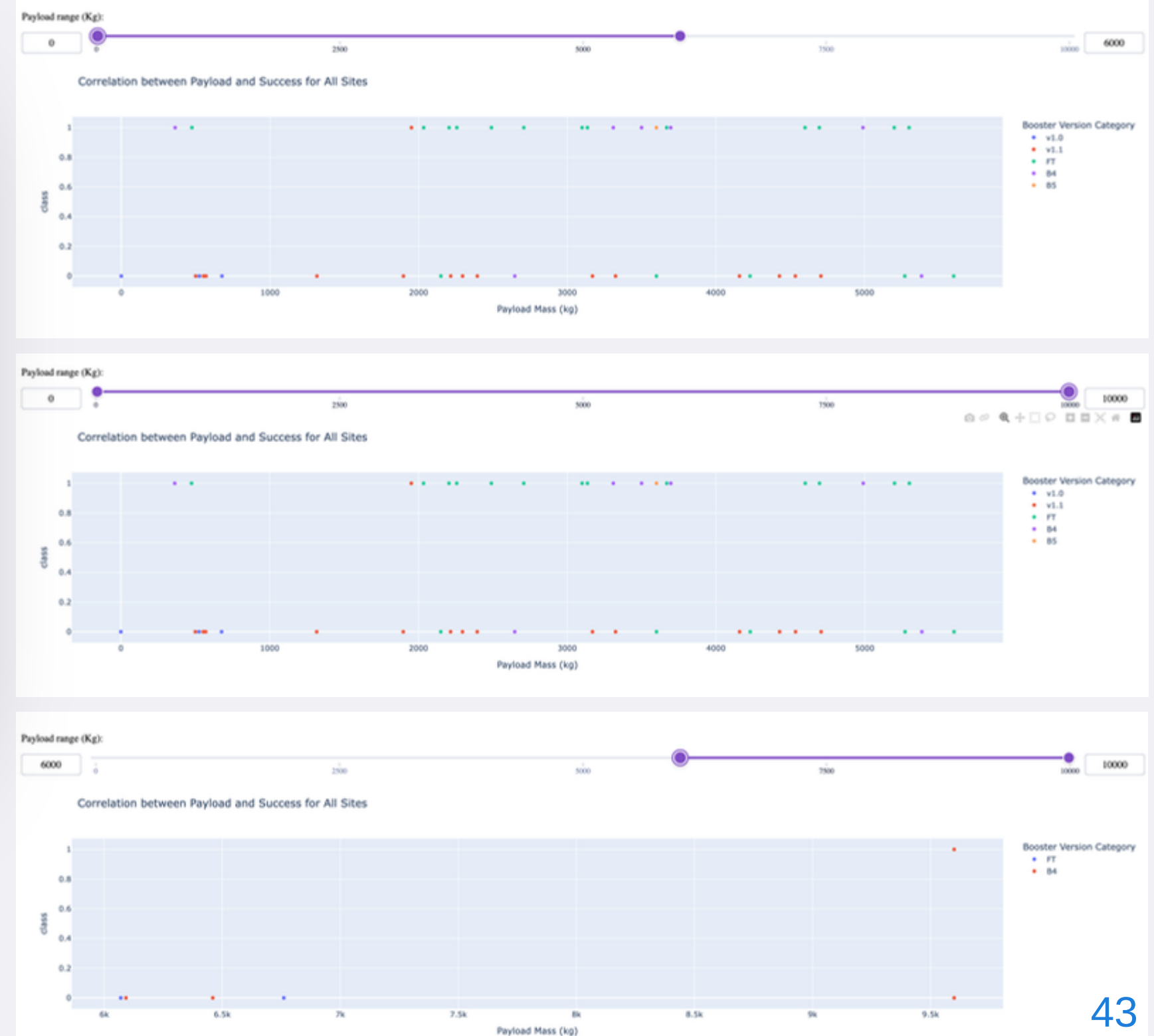
Inference:

- The launch site KSC LC-39A has a launch success rate of 76.9% .

Pay Load Mass and Success for All Sites

Inference:

- Payload range of 6000 kg and below has the highest launch success rate
- While a payload range of 6000 kg and above has the lowest launch success rate
- Given this graph, it can be analyzed that the booster version FT has the highest launch success rate



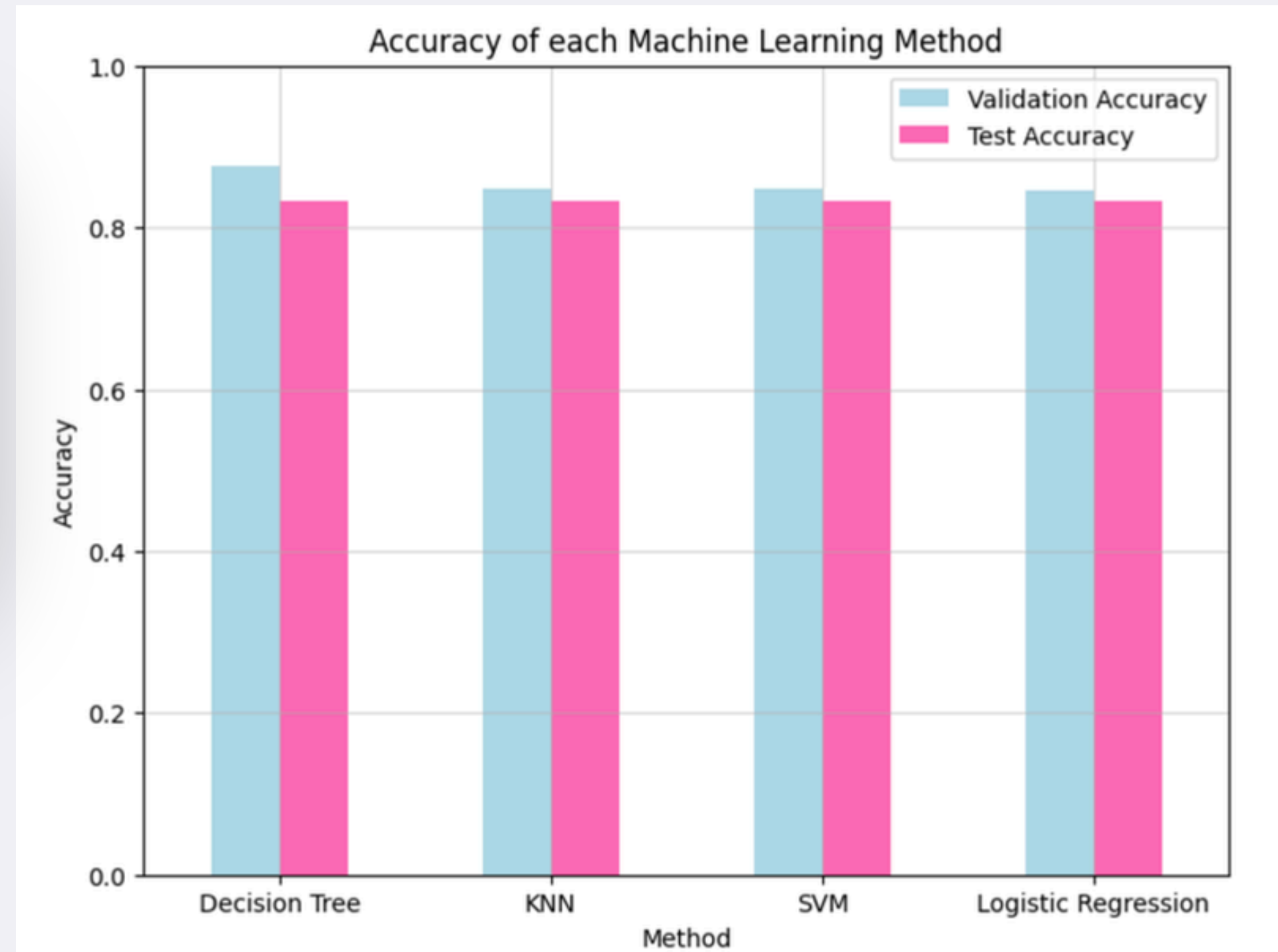
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Inference:

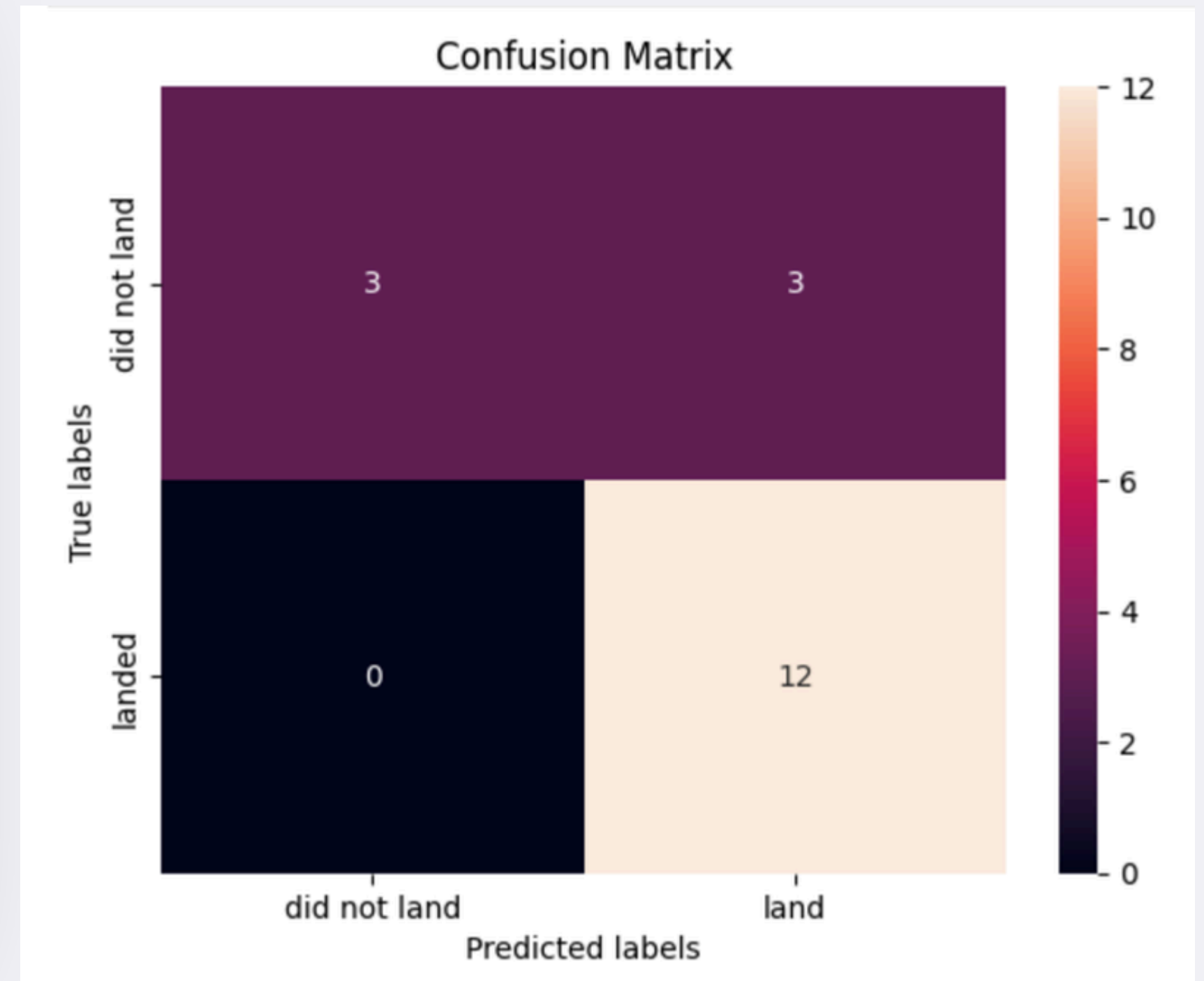
- Based on the accuracy on the test set and the validation set, the method that performed best was the **Decision Tree Classifier**



Confusion Matrix

Inference:

- The confusion matrix of the Decision Tree Classifier revealed the following:
 - False Positives (3)
 - False Negatives (0)
 - True Positives (12)
 - True Negatives (3)
- The amount of true positives are relatively higher and the false positives are lower, indicating an accuracy of 83.33%



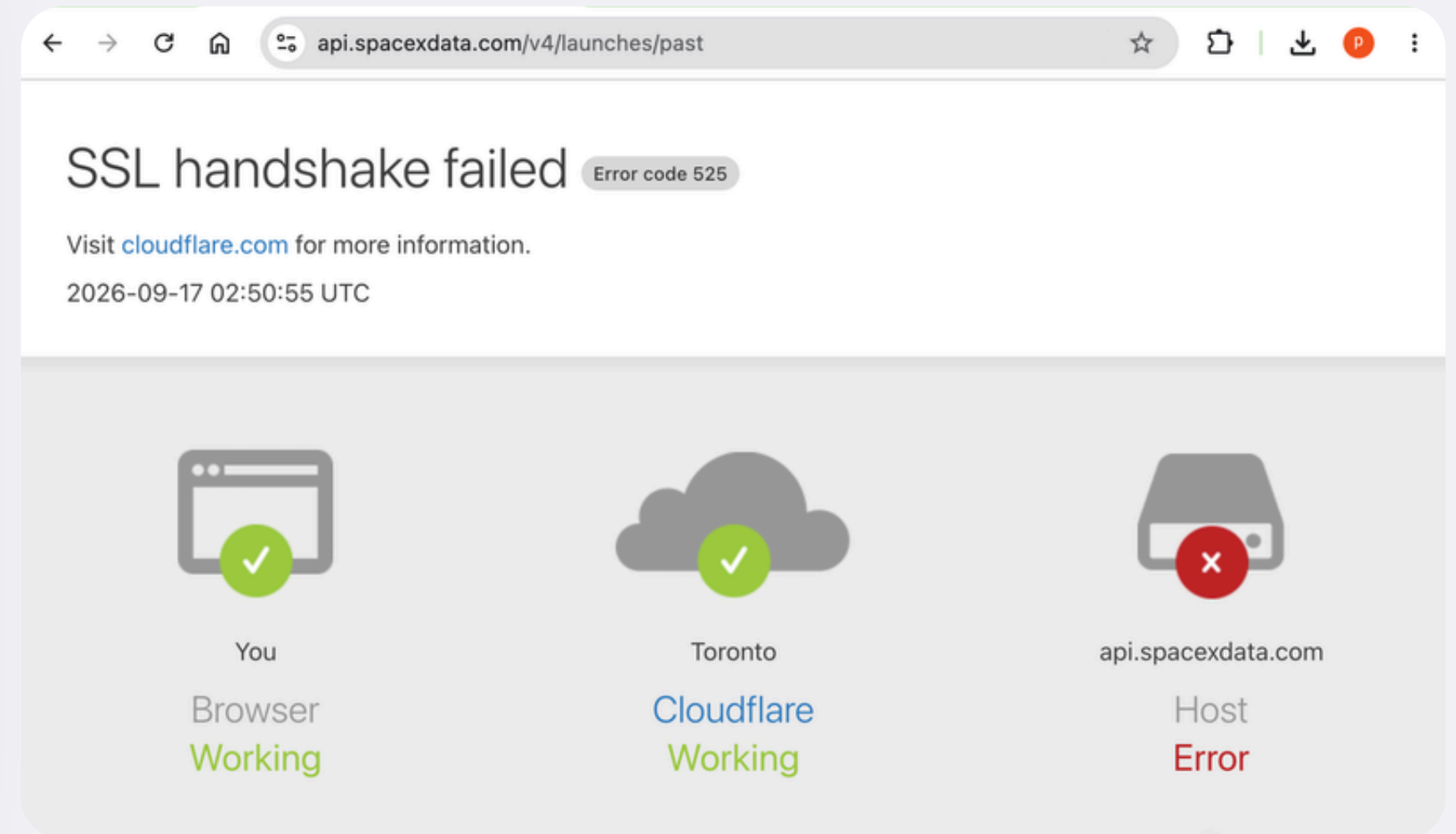
Conclusions

The analysis of Space X data revealed that several factors are linked to a successful rocket launch, specifically with the Falcon 9 launches:

- Orbit Types VLEO, ES-L1, GEO, HEO, SSO has the most amount of successful launches
- Overall, the success rate of launches has been increasing with time and number of flights
- Launch sites are situated near the coastline and the equator for both logistics and safety reasons
- Launch site KSC LC-39A has been the most successful launch site with a success rate of 76.9%
- A payload range below 6000 kg has shown to have the most successful launches
- The decision tree classifier is the best performing classification method with an 83.33% test accuracy, that can predict the landing outcome of a rocket launch.

Appendix

- The API link for the Space X launch data is non-existent and shows a 525 error, the code cells cannot call the functions, hence data cannot be fetched from the API.
- Still, the codes cells for the respective jupyter notebook are filled and completed even when they show an error when executed.



Thank you!

